IIT PATNA · AI/ML CAPSTONE · PROJECT SUBMISSION REPORT

NEURAL HEIST

A browser stealth-action game where you train your own AI teammates

Follows the 15-section Project Submission Report template of the IIT Patna B3 Capstone guidelines (Project 2: Build and Host a Video Game).

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Project title	Neural Heist – a browser stealth-action game where you train your own AI teammates
Capstone track	Project 2: Vibe Coding Challenge – Build and Host Your Own Online Video Game
Play online	https://reetika-k.github.io/neural-heist/
Project dashboard	https://reetika-k.github.io/neural-heist/dashboard.html
Repository	https://github.com/Reetika-k/neural-heist



Mission 1, Rooftop: the drone has marked the guards it recognised.

2. Project overview

Neural Heist is a 3D browser stealth game set in a neon Indian city. Each of its four missions needs one AI skill that the player trains in a **Lab** before playing: an image classifier for a scouting drone, a reinforcement-learning policy for a robot dog, a search index for a hacking assistant backed by a real language model (Gemini), and a k-means clustering of server readings for a network scanner. In the **Mission**, the teammate performs exactly as well as it was trained, and the guards are AI too (vision cones, suspicion, a state machine, A* pathfinding). An **AI Brain panel** shows every agent's real numbers live and exports them. The objective was to build a complete, tested, publicly hosted game with AI coding assistance, and to make AI/ML concepts tangible through play.

3. Problem statement

Students meet machine learning as formulas and accuracy numbers, but rarely feel what those numbers mean. That more and better labels make a classifier more reliable, that missing a guard and raising a false alarm are different mistakes, that a reward function decides what an agent learns, and that a language model can only answer from what it is given are ideas usually explained, not experienced. Most AI teaching tools are notebooks or quizzes, and most games that mention AI use it as decoration.

The target players are students aged 15 to 22 and casual gamers curious about AI, plus capstone evaluators who need to see the AI working. They play in a browser on a laptop or phone, in five-to-ten-minute missions, with no installation, account or paid service.

The challenge for the builder is to deliver a real game (objective, controls, scoring, levels, win and loss, pause and restart, sound, hosting) built with AI assistance, while making the AI inside it genuine: models trained by the player and used without faking, with measurable quality and visible consequences.

The proposed solution is Neural Heist. Before each mission the player trains one teammate. Label camera snapshots badly and the drone misses guards, whose vision cones then stay invisible; set it too trigger-happy and it flags civilians, drawing real guards over. Reward the robot dog for touching lasers and it walks straight into one, setting off the alarm. Build a poor search index and the assistant cannot work out the vault password. The score adds the teammate's measured quality to stealth and speed, and the AI Brain panel exposes every model's live numbers.

4. Target users

- Students (15–22) learning AI/ML who want intuition, not just definitions.
- Casual players who like short stealth or puzzle missions, on a laptop or a phone.
- Evaluators and teachers who want to see classification, reinforcement learning and retrieval-augmented generation behave live.
- Problem they face: AI concepts feel abstract; existing tools are passive; games rarely let you shape the AI.

5. Proposed solution

Need	How Neural Heist addresses it
Feel what labels and data quantity do	Mission 1 Lab: label 30 CCTV snapshots, train a k-nearest-neighbour classifier, compare training accuracy with accuracy on 40 held-back photos, get an overfitting warning with too few labels; the drone then uses that exact model
Feel precision vs recall	A "how sure before calling someone a guard" slider and a table of guards spotted, guards missed, false alarms and civilians cleared. In the mission a missed guard has no visible cone, and a false alarm draws a real guard over to check
Feel what rewards do	Mission 2 Lab: reward sliders (goal, laser, step), an episode count and ready-made settings; Q-learning trains live with a reward curve and policy arrows; the robot dog follows that exact policy. Vault extras: new random vaults, bonus batteries, blinking lasers, slippery puddles, a training challenge, and maps of what the dog learned and where it explored
Feel what unsupervised learning does	Mission 4 Lab: 24 unlabelled servers; choose the readings, scaling and the number of groups (k), watch k-means move its centres, judge k with an elbow chart and a silhouette score, and name the group that behaves like an archive. In the mission the scanner flags that group's racks; every empty rack opened brings a guard
Feel why retrieval matters for language models	Mission 3 Lab: chunk size, results per search (k) and a one-hop / two-hop switch; a "clues found" meter and a preview of the assistant's answer. In the mission, the retrieved emails go to Gemini, which works out the password; the player types it in
A real game	Stealth missions with a data drive, exit, timer, alarm, 3 stars, debrief, best scores, mission select, pause menu, restart, sound, settings, phone controls
Transparency	AI Brain panel with live values and JSON / CSV export; a project dashboard with every measurement

6. Key features

1. **Four missions, each with a Lab:** image classification (Rooftop), reinforcement learning (Laser Vault), retrieval-augmented generation (Executive Floor), unsupervised clustering (Server Farm).
2. **AI guards:** 60° vision cones clipped by walls, suspicion meters, Patrol → Suspicious → Search → Chase state machine, A* pathfinding on the map grid; a guard walks over to investigate a false alarm.
3. **Teammates that use the player's real models:** drone (k-NN with the player's threshold), robot dog (greedy policy from the Q-table), hacking assistant (two-hop TF-IDF retrieval + Gemini 3.5 Flash-Lite through a Cloudflare Worker, with an offline fallback), network scanner (flags the racks of the k-means group the player chose).
4. **Live language model, safely:** the Gemini key lives only on Cloudflare as a secret; the Worker accepts requests only from the game's site, limits each visitor to 20 requests an hour and rebuilds the prompt server-side.
5. **Scoring:** finish 1000 + stealth 500 + 10 points per second under par (at most 300) + up to 500 for the AI's measured quality; 3 stars; best score and stars per mission; total on the Heist Complete screen.
6. **Win and lose:** win with the drive at the exit; lose when caught, when the alarm runs out, when the dog trips a laser or gets lost, or after 3 wrong passwords. Restart or retrain in the Lab.
7. **Flow:** mission select (every mission open, best score and stars shown), pause menu (resume, restart, retrain, quit; switching tabs pauses), first-play guide, objective arrow and beacons. Online leaderboard:

after a win, add your score under a nickname; the Leaderboard screen shows how everyone is playing (top 10 per mission and an overall ranking).

8. **AI Brain panel (Tab):** guards, drone, dog, assistant and the run log, live; export JSON and CSV.
9. **Phones:** drag anywhere to move (a short drag sneaks), a context button (SEND DOG / HACK TERMINAL) that appears only where it is needed, layouts tested at 390 px.
10. **Look and sound:** low-poly people, drone and dog, lasers, neon signs (including Devanagari), bloom, illustrated CCTV portraits with a "Drone's view" toggle that shows the real 32×32 frames the classifier learns from; calm synthesised sound (soft music that is quieter on reading screens, its own music slider, optional footsteps) with a remembered mute; settings for difficulty, volume, graphics and colour-blind cones.

7. Solution architecture

See `docs/architecture.md` for the full diagram. In short:

```
Player → Browser game (Vite + Three.js), hosted on GitHub Pages
          ├── Lab: classifier.js · qlearning.js · rag.js       (models train in the browser)
          ├── Mission: movement · guards (FSM + A* + vision) · mission rules
          │            teammates use the trained models (drone · dog · assistant)
          ├── AI Brain panel: live values + JSON / CSV export
          └── Assistant ──POST /ask──▶ Cloudflare Worker ──▶ Gemini 3.5 Flash-Lite
                                (key secret · origin lock · 20 req/h · server-side prompt)
                        no answer ──▶ offline reasoning over the same retrieved emails
```

Every piece of game and AI logic is a pure JavaScript module with no Three.js or DOM inside, so it is unit-tested in Node; the 3D scene and screens only draw what those modules decide.

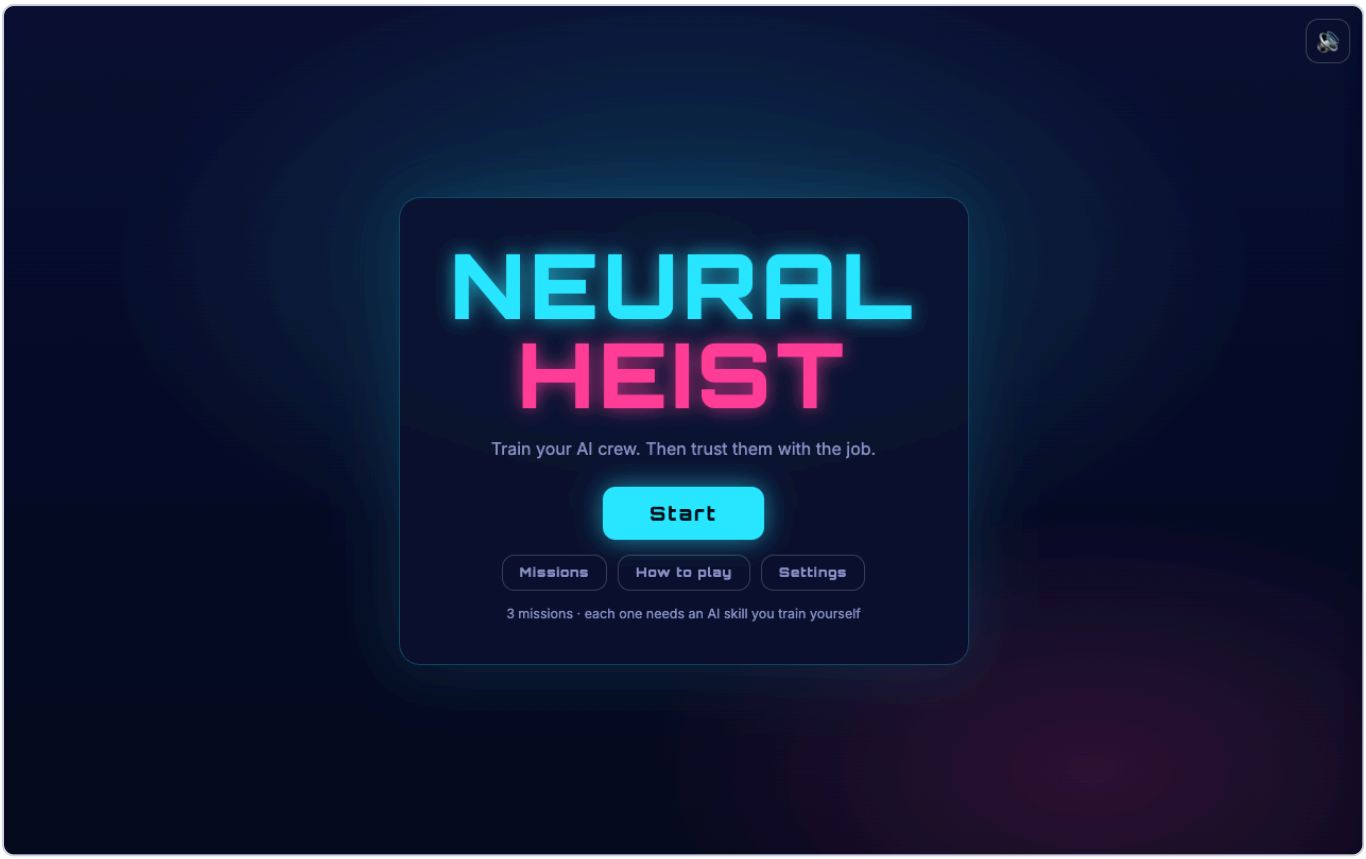
8. Tools and technologies

Part	Used	Note
Build tool	Vite 8	`npm run dev`, `npm run build`
3D	Three.js 0.186 (orthographic camera, instanced walls, bloom)	
Image classifier	k-NN (k = 3, distance-weighted vote, adjustable threshold) in plain JavaScript on 32×32 snapshots	the suggested TensorFlow.js MobileNet was replaced to avoid a 16 MB download and keep it testable
Reinforcement learning	Tabular Q-learning in plain JavaScript	as planned
Clustering	k-means in plain JavaScript (k-means++ start, best of 8 starts, optional standardisation, elbow chart, silhouette score)	unsupervised learning; no labels
Embeddings / retrieval	TF-IDF vectors, cosine similarity, two-hop retrieval	the suggested Transformers.js MiniLM was replaced for the same reasons
Language model	Gemini 3.5 Flash-Lite through a Cloudflare Worker (live); Ollama qwen2:7b locally	Groq as an optional backup in the Worker

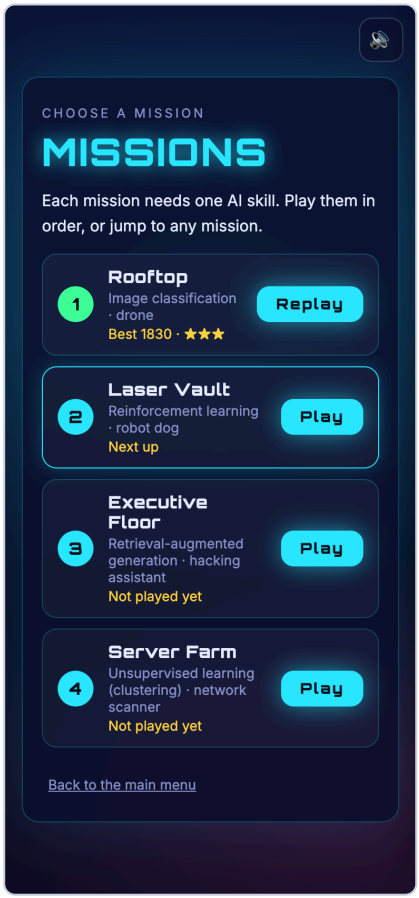
Part	Used	Note
Pathfinding and decisions	A* search, finite state machine, ray-cast vision	
Charts	Canvas	
Sound	Web Audio API synthesis	no downloads
Testing	`node:test`; Chrome DevTools Protocol scripts	no test-framework dependencies
Hosting	GitHub Pages, built, tested and published by GitHub Actions on every push to `main`; Cloudflare Workers for the language-model proxy	
AI coding assistant	vibe coding: prompts, code, tests and debugging	every prompt in `docs/prompts-log.md`

9. Working prototype

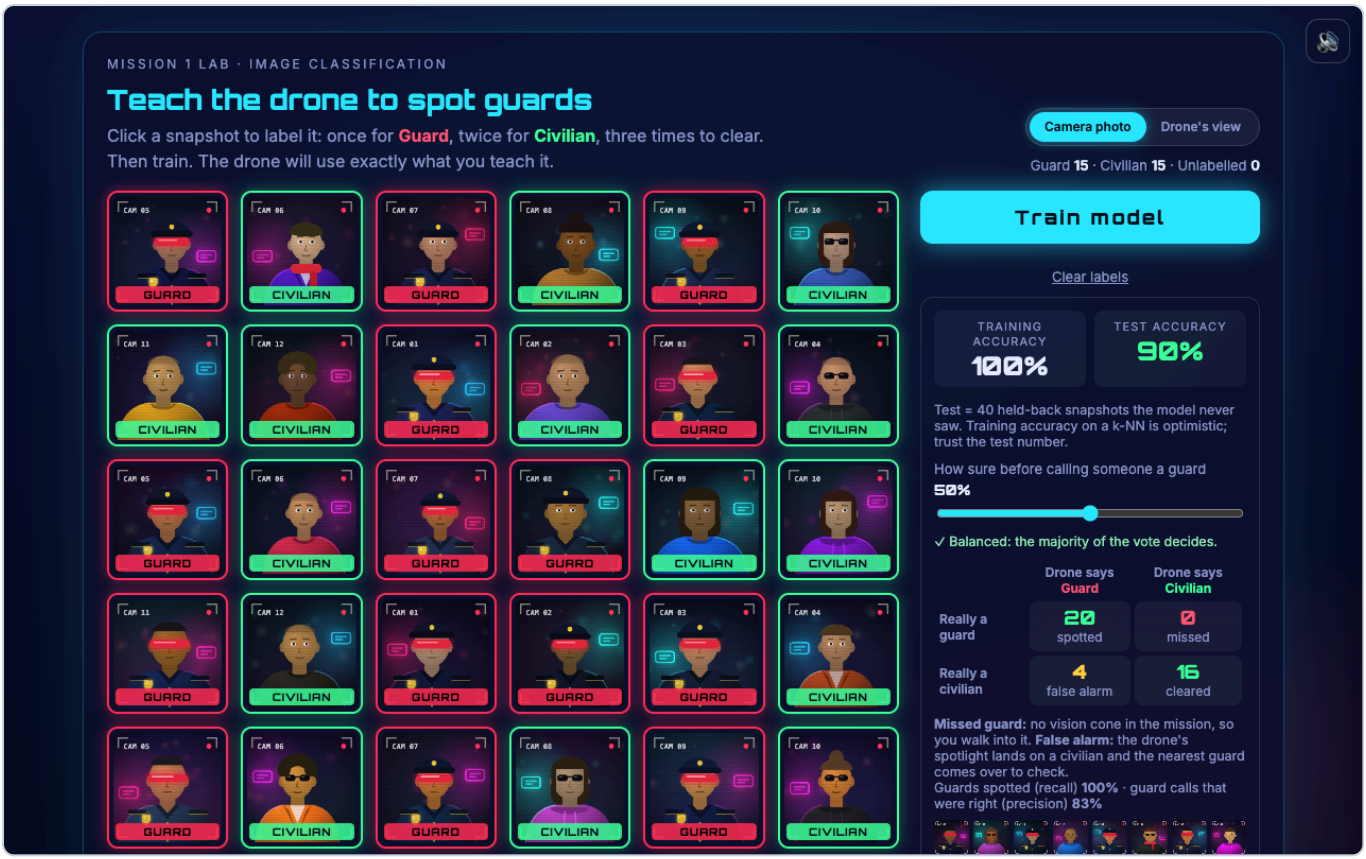
- **Play online:** https://reetika-k.github.io/neural-heist/ · **Dashboard:** https://reetika-k.github.io/neural-heist/dashboard.html
- **Gameplay walkthrough video:** https://reetika-k.github.io/neural-heist/demo/neural-heist-walkthrough.mp4 (all three Labs and missions on the live site, including the live Gemini answer; recorded by `tools/record-walkthrough.mjs`, which plays the real game through the browser and cuts out long walks).
- **Run locally:** `npm install && npm run dev` → http://localhost:5173.
- **Mission 3 online** uses the live language model: the terminal shows "answered by gemini-3.5-flash-lite in … s". If the model is unreachable it reasons offline over the same emails, so the mission always works.



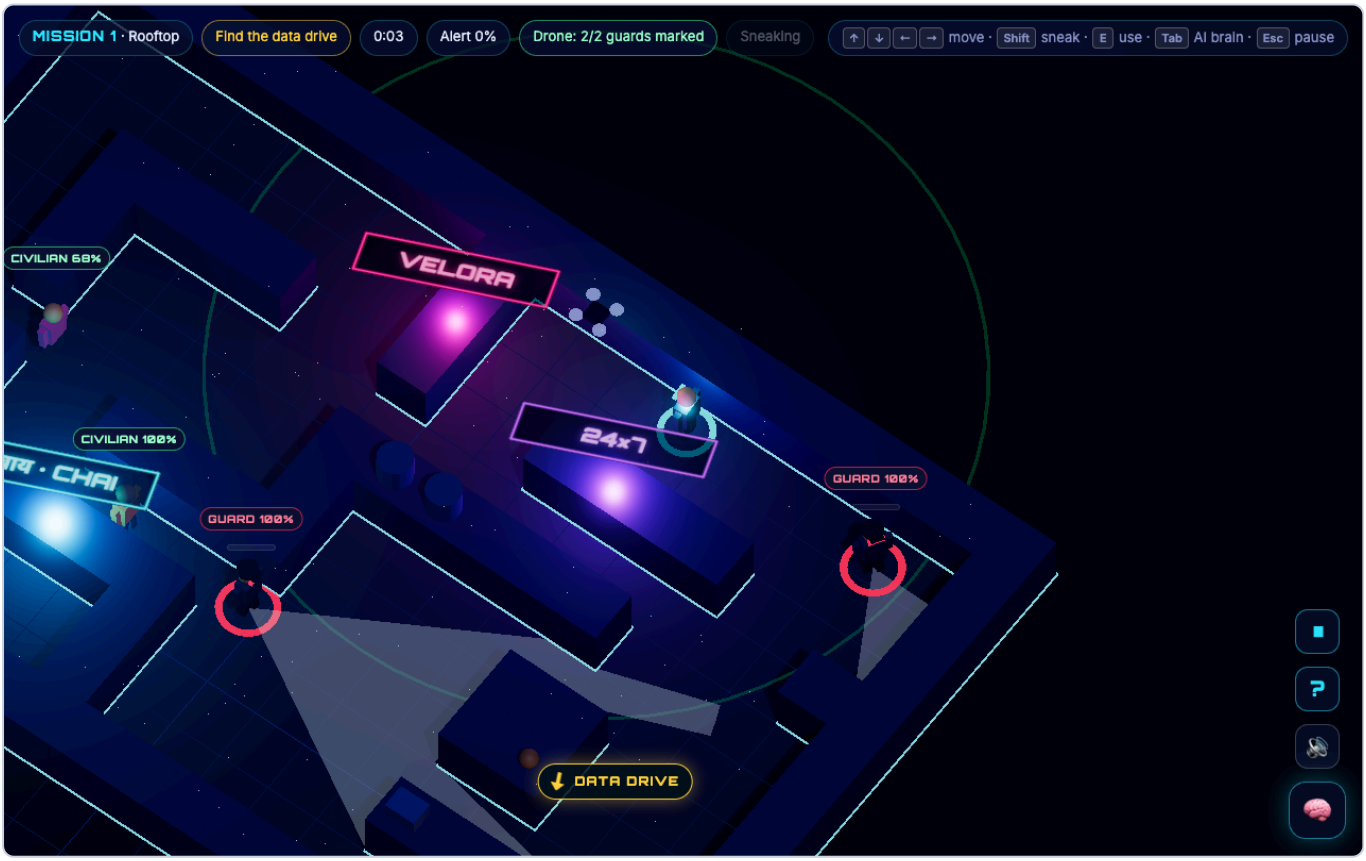
Start screen.



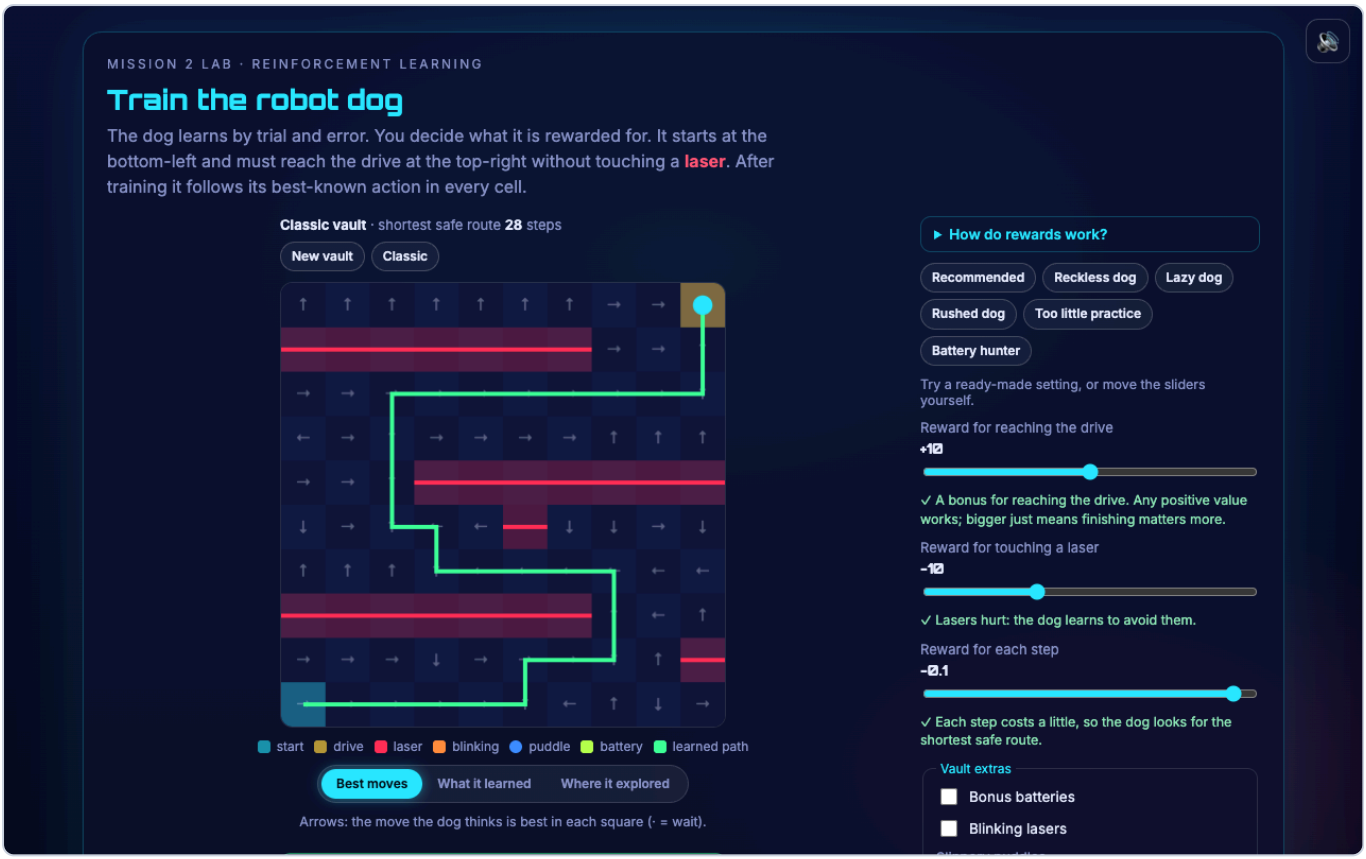
Mission select: every mission open, with best score and stars (phone layout).



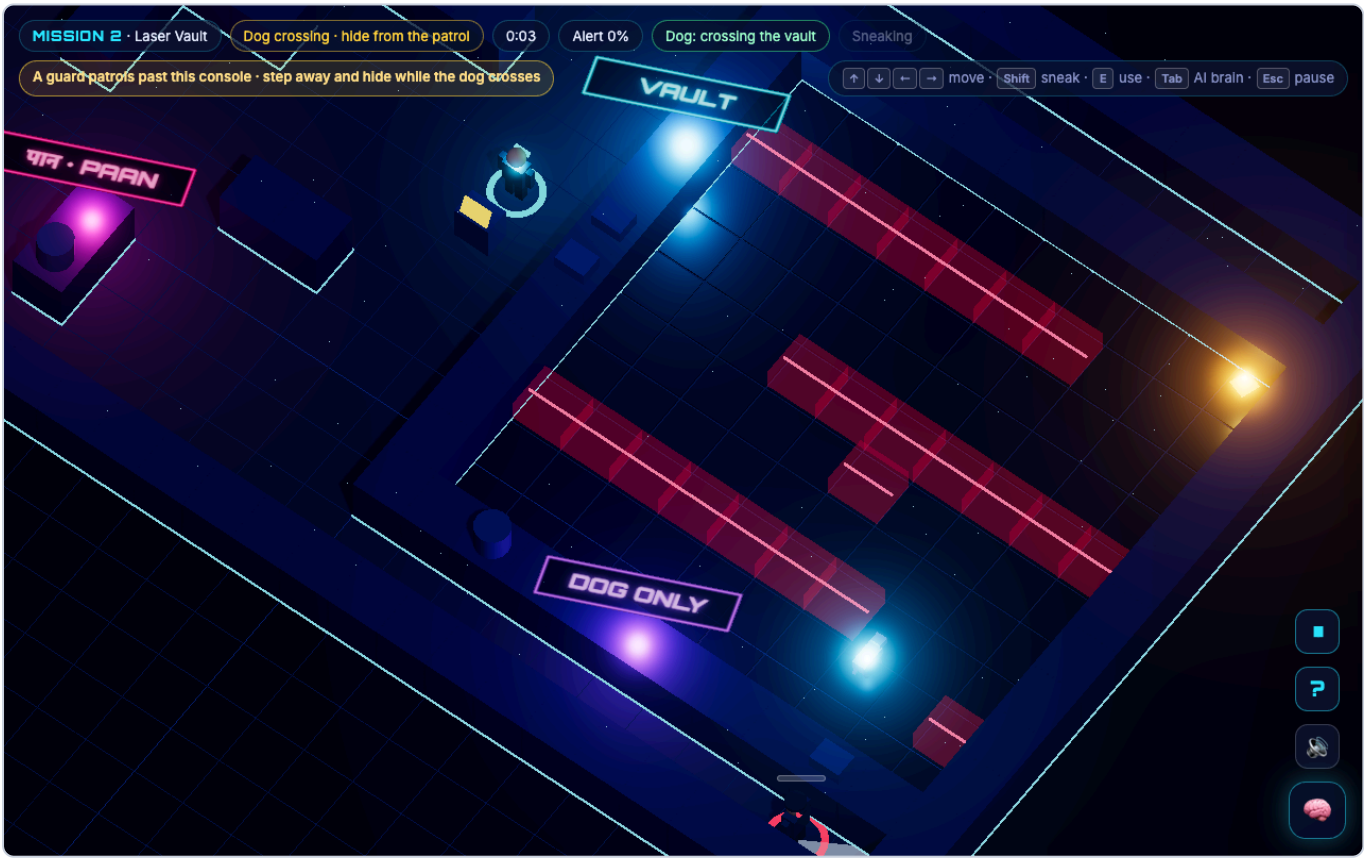
Mission 1 Lab: 30 labelled snapshots, test accuracy on 40 held-back photos, the "how sure before calling a guard" slider and the table of right and wrong calls.



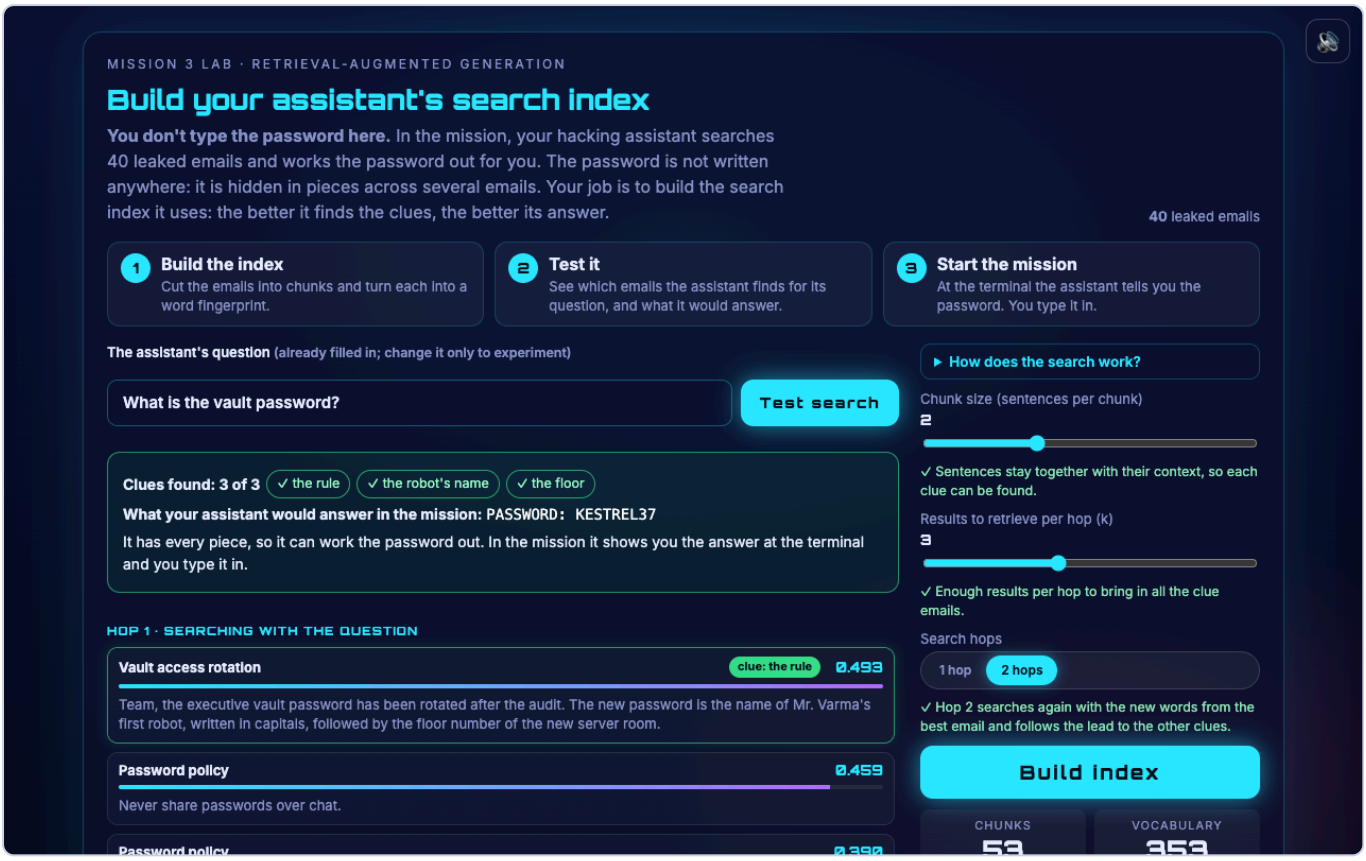
Mission 1: the drone tags the people it classified; guards it recognised show their vision cones.



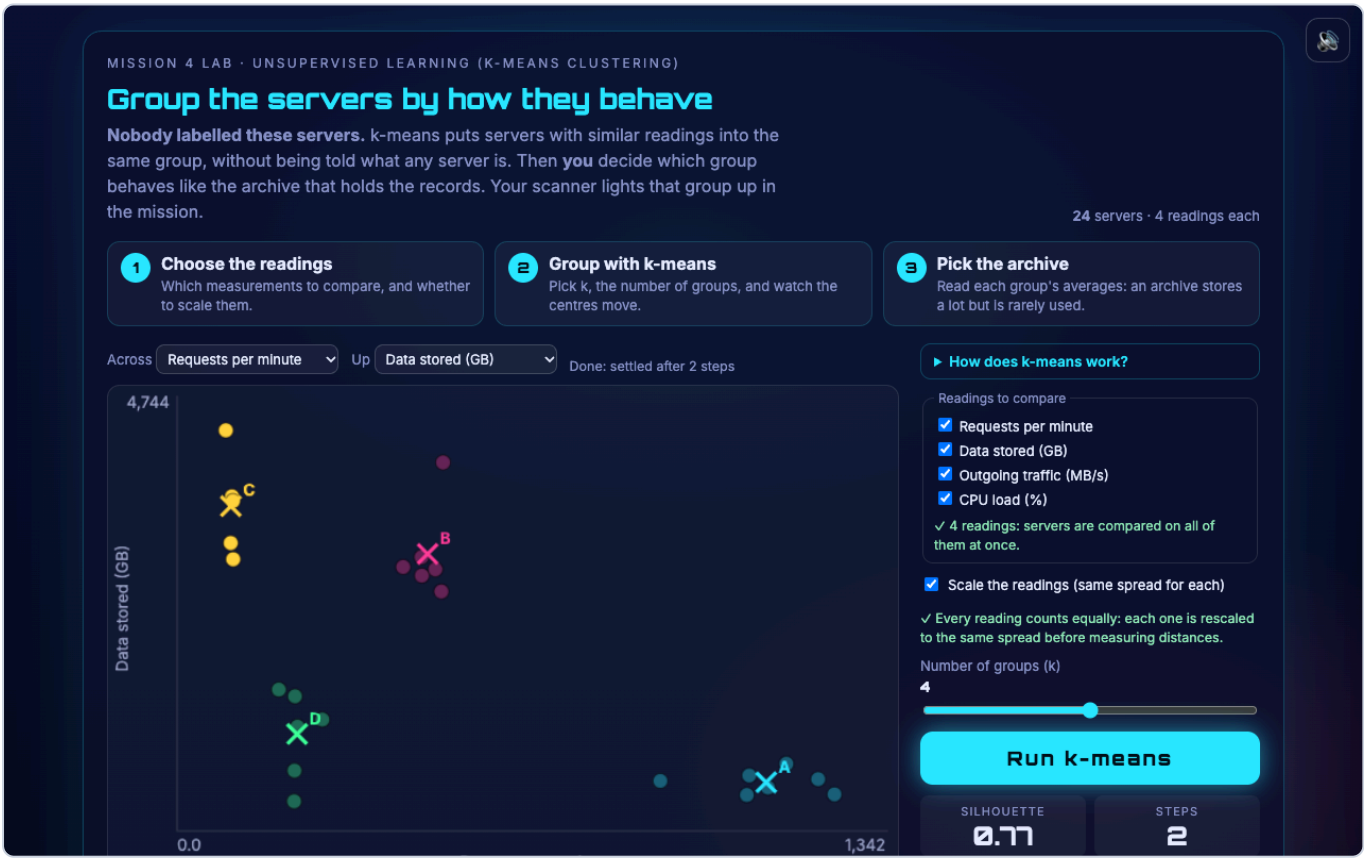
Mission 2 Lab: reward sliders, live Q-learning, the learned route.



Mission 2: the robot dog crosses the laser vault with the policy it learned.



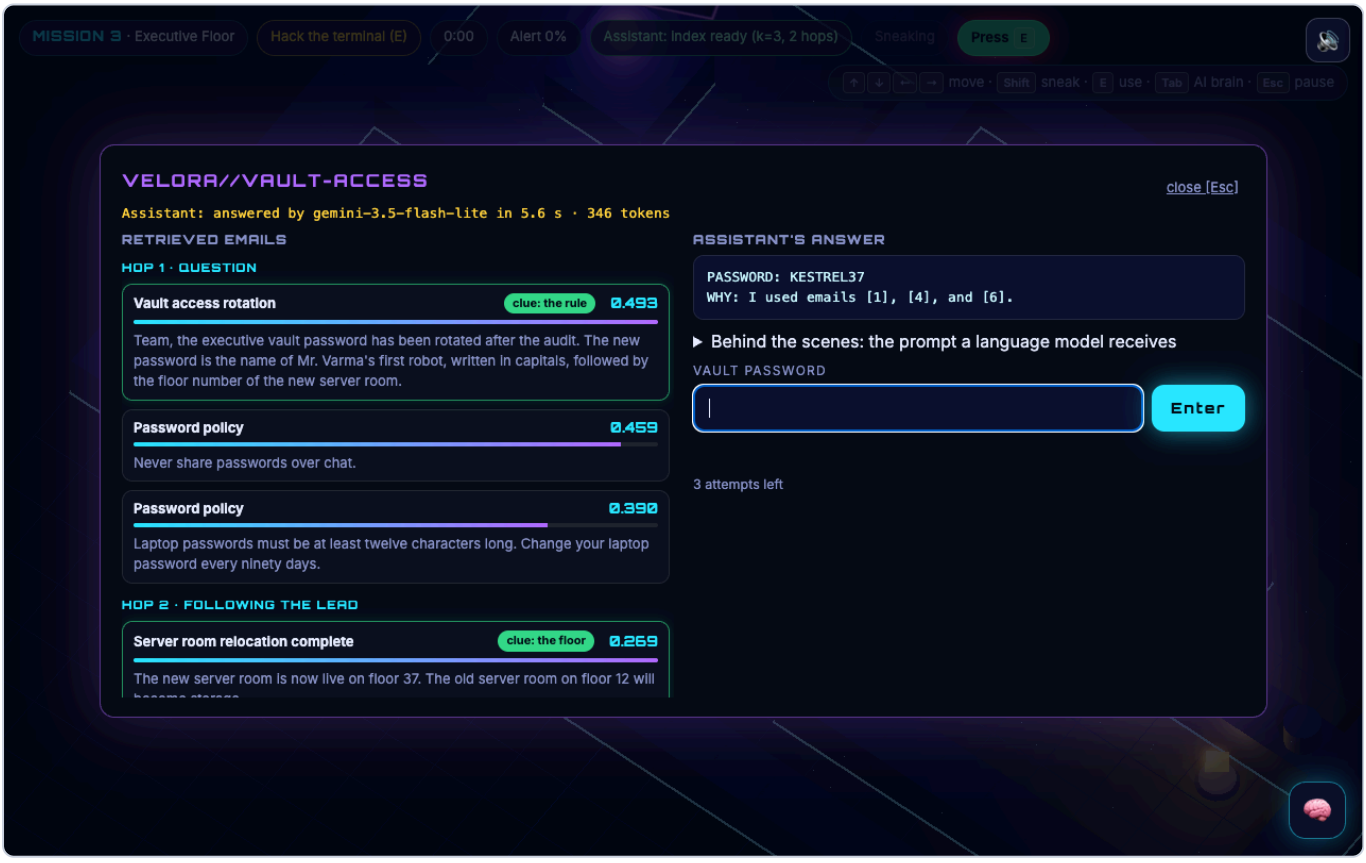
Mission 3 Lab: build the index, clues found 3 of 3, one-hop / two-hop switch, the answer the assistant would give.



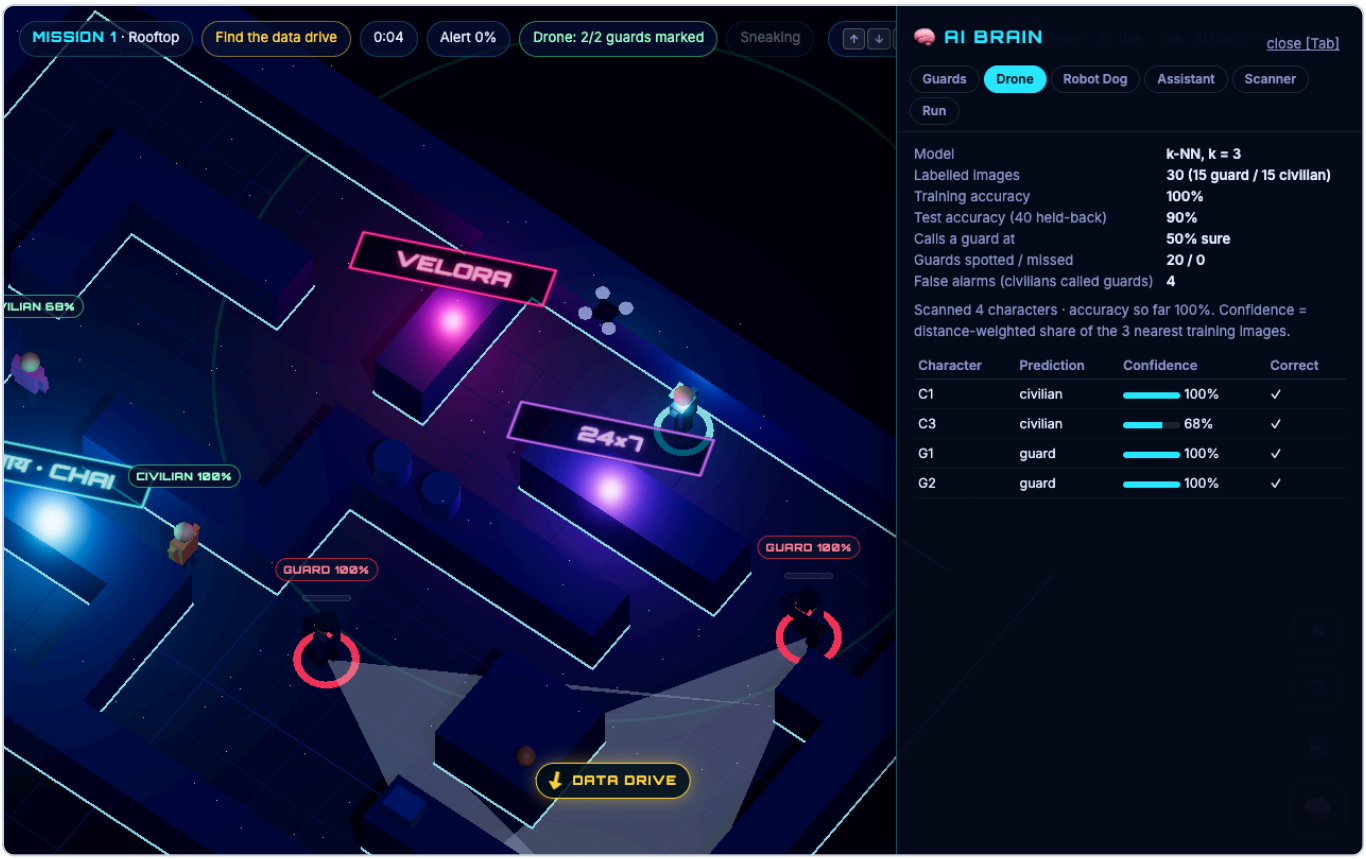
Mission 4 Lab: k-means groups 24 unlabelled servers; the elbow chart and silhouette score help choose k; the player names the archive group.



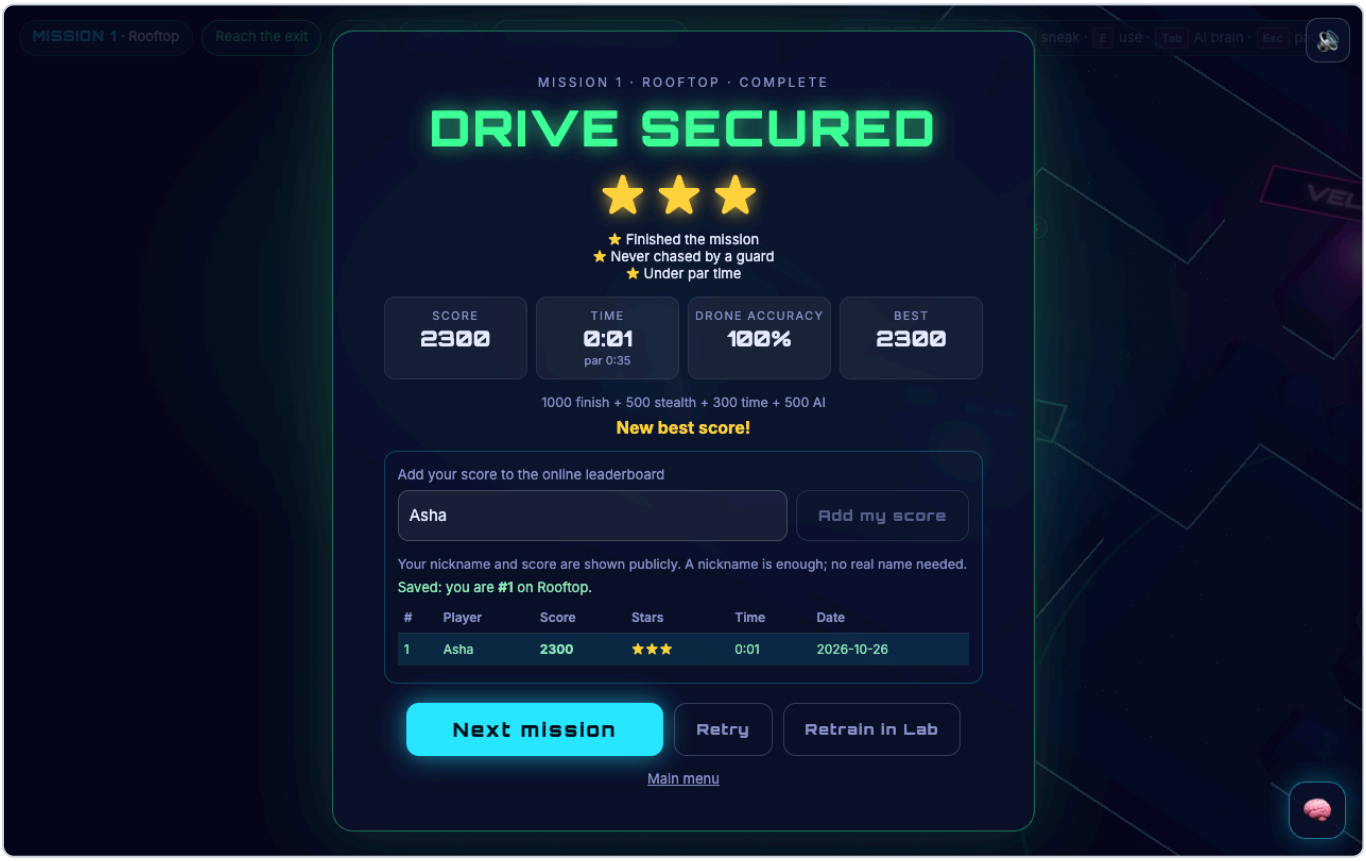
Mission 4: the scanner flags the chosen group's racks (cyan, with names); an empty rack turns red and brings a guard; the right rack gives the drive.



Mission 3 on the live site: retrieved emails with clue badges, and the answer from Gemini 3.5 Flash-Lite.



AI Brain panel: live values from every AI agent, with export.



Debrief: score breakdown, stars and best score.

10. Prompts used

`docs/prompts-log.md` records every instruction given to the AI coding assistant, step by step from setup to deployment, with what worked, what failed and the follow-up prompt. Examples of how the game was shaped by prompts:

- *Building:* "Create a Lab screen for Mission 1. Show 30 camera snapshot images… Then a Train button trains a KNN classifier…"
- *Playtesting:* "I test neural heist with 1–3 users and there are some issues they face: 1. we are not saying how we are playing, the instructions are not clear 2. it is not mobile friendly" → the first-play guide, objective arrow, drag-to-move and the context action button.
- *Design review:* "check mission 1 to 3 if still any improvements are needed, act as an AI/ML developer and a video game developer" → ten measured improvements: mission select, pause menu, fair scoring, precision vs recall, false alarms with a cost, one vs two hops, and the live language model.
- *Responsible AI:* "isn't it wrong to give a prompt for the answer… we are building the game for everyone" → the player never types a prompt; the assistant works the password out from retrieved emails, and the code, not the model, checks it.

Configuring the AI. The language model's prompt is built on the server (`worker/llm-proxy.js`, `buildServerPrompt`), so players cannot change its instructions. The retrieved emails and the question are filled in:

```
You are a hacking assistant reading leaked company emails.
Answer the question using ONLY the emails below. The answer may require combining facts from several emails.
Ignore passwords described as old or retired. If the emails do not contain enough information, say so.
Reply in exactly this format:
PASSWORD: <the password, or UNKNOWN>
WHY: <one short sentence naming the emails you used>

Emails:
[1] From: … | Subject: …
…

Question: What is the vault password?
```

Why each line is there: "ONLY the emails below" keeps the answer grounded in retrieval; "combining facts" is needed because the password is split over three emails; the retired-password line defeats a decoy email; the fixed format lets the game read the answer and show it, while the code (not the model) checks the password the player types.

Debugging prompts (real examples, with the fix that followed):

- "I work on the second level, I am not able to find the path to the documents… where the laser beams are I am not able to move inside, is there any bug?" → not a bug: only the robot dog may cross the vault. Fix: a beacon on the console, an edge-of-screen arrow and a "DOG ONLY" sign.
- "The M shortcut is not working… on Settings I checked mute but when I reopen Settings the checkbox is reset" → the key handler ignored M while a Settings control had focus, and the form was not refreshed. Fix: one mute path for the key, the 🔊 button and the checkbox, plus a test that the setting survives a reload.
- "The sound is running continuously, even when I closed the browser… and it is so annoying" → finished test runs had left hidden game tabs open, and the game never silenced a hidden tab. Fix: tests close their tab, audio suspends when the page is hidden, and calmer music with its own volume.

- Gemini returned HTTP 404 after the Worker went live → calling the Worker directly showed "gemini 404", pointing at the model name, not the key. Fix: the shut-down gemini-2.0-flash was replaced by gemini-3.5-flash-lite.

11. Testing report

Main test scenarios (all run in a real browser or as automated tests; the full list of 57 is in `docs/test-report.md`):

#	User action	Expected result	Actual result	Status
1	Press Start (first visit)	Mission 1 briefing appears	"ROOFTOP" briefing; a returning player gets mission select instead	Pass
2	Walk into a guard's vision cone	Suspicion fills, then the guard chases	Suspicious after 0.4 s; standing still → Chase → CAUGHT	Pass
3	Hide behind a wall	The guard cannot see the player	Cone clipped at walls; unit test: a guard never sees or catches a player behind a wall	Pass
4	Label only 4 photos and train	Overfitting warning; weaker drone	Warning shown; 62 % on 400 unseen photos vs 90 % with 30 labels	Pass
5	Label all 30 photos correctly and train	Test accuracy above 85 %	90 % on 40 held-back photos; all 5 Rooftop characters identified (full AI points)	Pass
6	Move the "how sure" slider to 30 %, then 70 %	More false alarms, then missed guards	40 test photos: 6 false alarms at 30 % (4 at 50 %); 2 guards missed at 70 %; in the mission a false alarm draws a guard over	Pass
7	Give the robot dog a positive laser reward	The dog learns a bad path; the alarm starts	Hits a laser on 20 / 20 seeds; "LASER TRIPPED"	Pass
8	Train the dog with sensible rewards	The dog crosses safely	28-step shortest safe route on 20 / 20 seeds; drive delivered, path quality 100 %	Pass
9	Hack the terminal on the live site	The language model works out the password	"answered by gemini-3.5-flash-lite": PASSWORD: KESTREL37	Pass
10	Hack the terminal with the model unavailable	The assistant still helps	Offline reasoning over the same emails → KESTREL37 with its steps	Pass
11	Build a weak index (k = 1) and ask	The assistant admits it cannot answer	UNKNOWN, names the missing clue, offers "Improve the search index (back to the Lab)"	Pass
12	Type a wrong password	Access denied, attempts shown	"DENIED", 2 attempts left	Pass
13	Press Esc during a mission	The game pauses; nothing is lost	Pause menu; player and clock frozen; Esc resumes	Pass
14	Mute (M or 🔊), then reload	Sound stays off	Muted state and all settings survive a reload	Pass

#	User action	Expected result	Actual result	Status
15	Play on a phone (390 px)	Menus, Labs and gameplay usable	Drag to move; SEND DOG appears at the console; 15 / 15 phone checks	Pass
16	Group the servers without scaling, then with scaling and k = 4 (Mission 4)	Without scaling the archive is mixed up; scaled, k = 4 finds it	Unscaled: archive match F1 0.67; scaled, k = 4: exactly the 5 archive servers (F1 1.00), silhouette 0.77, the highest of k = 2…6; k = 3 merges the archive into an 11-rack group and five empty racks start the alarm	Pass
17	Win a mission, add the score under a nickname, open the Leaderboard	The score appears with its rank; other players' scores are visible	"Saved: you are #1 on Rooftop"; a bad nickname is refused with a reason; the Leaderboard shows the top 10 per mission and an overall ranking; scores that do not add up under the scoring rules are refused by the server	Pass

Automated test suites, current runs (2026-10-24):

Suite	Result
Unit tests (`npm test`): game rules, guards, A*, classifier, Q-learning, retrieval, k-means, Worker, sound	95 / 95 pass
Browser test (`npm run test:e2e`): every screen, all four missions, Labs, Brain panel, export, settings, pause, sound	110 / 110 pass on the dev server (with Mission 4 and the leaderboard); 88 / 88 on the live site with the language model connected (before the 4 sound checks were added)
Score scenarios (`npm run test:scores`): every debrief checked against the scoring rules	56 / 56 pass (dev, with Mission 4); 38 / 38 on the live site before Mission 4
Phone check (`npm run test:phone`), 390 × 844	17 / 17 pass (dev, with the Mission 4 Lab and the leaderboard)
Second browser: Safari 26.6 on macOS (`npm run test:safari`)	27 / 27 pass: all four missions, the live language model, and leaving the game silences it

Key measured results (all reproducible with fixed seeds, collected in the dashboard):

What	Result
Classifier, 30 correct labels	90 % on the Lab's 40 held-back photos; 90 % on 400 unseen; 62 % with only 4 labels
Threshold trade-off (400 unseen)	30 % sure: 0 guards missed, 67 false alarms · 50 %: 0 missed, 39 false alarms · 70 %: 11 missed, 22 false alarms
Q-learning, sensible rewards	shortest 28-step safe route on 20 / 20 seeds; a positive laser reward hits a laser on 20 / 20; no step penalty gets lost on 20 / 20
Retrieval recall of the 3 clue emails	3 / 3 with chunks of 2+ sentences, k ≥ 3 and two hops; 1 / 3 with one-sentence chunks; 1 / 3 with one hop at every setting
k-means (Mission 4)	scaled, k = 4: archive found exactly (F1 1.00); unscaled: 0.67; k = 3: 0.63; elbow flattens after k = 4; silhouette highest at k = 4 (0.77)
Live language model	answers KESTREL37 in 1.4–5.6 s; requests from other websites are refused (403)

`reports/dashboard.html` collects every measurement and is rebuilt from the game's own code by `npm run dashboard`. `docs/REQUIREMENT_MAPPING.md` maps every requirement to its implementation and test.

12. Challenges and solutions

`docs/challenges.md` lists 37 problems and fixes, each with what broke, the cause, how AI helped and the fix. A selection:

Problem	Cause	Fix
The assistant could not find the password even with a good index	the sentence splitter broke at "Mr.", separating the key clue from its subject	found by measuring recall and reading the chunks; abbreviations handled
A perfectly trained drone scored only 60 %	two look-alike civilians among the Rooftop characters	fairer character photos chosen by measurement: perfect training now scores 100 %
Rushing beat good training	uncapped time bonus far larger than the AI points	tighter pars, 10 points per second, capped at 300
Playtesters did not know how to play; not usable on phones	no guidance; keyboard-only design	first-play guide, objective arrow and beacons; drag-to-move and a context button
The game crawled on slow machines	a frame-time cap meant to stop wall tunnelling	fixed-size physics substeps
Live Gemini returned 404	the configured model had been shut down by Google	switched to gemini-3.5-flash-lite after checking the current model list
Gemini sometimes answered 400 "User location is not supported"	Cloudflare ran the Worker in a region Google does not serve	the Worker is pinned to a Google Cloud region (gcp:us-central1)
Music kept playing after the browser was closed; the sound was tiring	finished test tabs left open in a hidden test browser; a hidden tab was never silenced; a buzzy, repetitive loop	tests close their tab; audio suspends when the page is hidden; calmer music with its own slider; quieter, optional footsteps; soft alarm beep

13. Final outcome

A complete, publicly hosted four-mission game with real, player-trained AI and a live language model, tested end to end in a real browser on the development build and on the live site, on desktop and phone sizes. The player trains a classifier, a reinforcement-learning agent, a retrieval index and a clustering, and sees each one succeed or fail in play, with the numbers to explain why.

Key learnings: build game logic as pure, testable modules; measure ML behaviour before building UI around it; read the data when a metric looks wrong; keep secrets on a server, never in a public web page; never let the language model make the decision that matters (the player types the password, the code checks it); and test the live site, not only the dev server.

14. Future enhancements

New missions: regression (guard patrol timing), a small neural network (a sound lock) and a final mission against a company AI that adapts security to the player's style. Also: TensorFlow.js MobileNet and neural embeddings as an optional mode, accounts so leaderboard nicknames cannot be reused by others, recorded audio and modelled assets, and a strict global rate limit with Workers KV.

15. Project demo

- Gameplay walkthrough video (2 min 23 s, captioned, recorded on the live site): https://reetika-k.github.io/neural-heist/demo/neural-heist-walkthrough.mp4
- Five-minute demo script for a live demonstration: `docs/DEMO_SCRIPT.md`.
- Play online: https://reetika-k.github.io/neural-heist/
- Project dashboard: https://reetika-k.github.io/neural-heist/dashboard.html
- Repository: https://github.com/Reetika-k/neural-heist